Domination polynomials of clique joins: two-blade ultra-log-concavity and three-blade counterexamples

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Abstract

For all positive integers r, a, b, the domination polynomial of $K_r \vee (K_a \sqcup K_b)$ is ultra-log-concave of order $N = r + a + b$, and hence log-concave and unimodal. The proof uses an exact second difference for normalized binomial coefficients, with the zero constant coefficient and all truncation endpoints treated explicitly. For three blades, ordinary log-concavity can fail: $K_1 \vee (K_4 \sqcup K_5 \sqcup K_7)$ has first coefficients $(d_1, d_2, d_3) = (1, 16, 260)$. When equal blade sizes are allowed, $K_1 \vee (K_q \sqcup K_q \sqcup K_q)$ fails log-concavity for every integer $q \geq 5$. These results address parts of Rather's question following Theorem 2.16; the seventeen-vertex example is itself unimodal, so it is not a counterexample to general unimodality.

1 Graphs, coefficient properties and the source question

All graphs are finite, simple and undirected. A set S of vertices dominates a graph if every vertex outside S has a neighbor in S. Write

$$D_G(x) = \sum_{k=0}^{N} d_k x^k, \qquad d_k = \#\{S \subseteq V(G) : |S| = k,\ S \text{ dominates } G\},$$

where $N = |V(G)|$. Every graph considered here is nonempty, so $d_0 = 0$.

For positive integers $r, t, a_1, \ldots, a_t$, consider

$$G = K_r \vee \left(\bigsqcup_{i=1}^{t} K_{a_i} \right), \qquad M = \sum_{i=1}^{t} a_i, \qquad N = r + M. \tag{1}$$

The join adds every edge from the central clique K_r to each blade; distinct blades have no edges between them. Blade orders may coincide in this definition. The explicit counterexample below has distinct blade orders $4, 5, 7$.

A nonnegative sequence $(d_k)_{k=0}^{N}$ is log-concave if $d_k^2 \geq d_{k-1} d_{k+1}$ for $1 \leq k \leq N - 1$. It is *ultra-log-concave of order N* if

$$u_k = \frac{d_k}{\binom{N}{k}}, \qquad 0 \leq k \leq N,$$

is log-concave and has no internal zeros. A sequence is unimodal if it is nondecreasing up to some index and nonincreasing thereafter; equal adjacent entries are allowed.

Rather asks about domination zeros and their limits, log-concavity and unimodality for this clique-join family immediately after Theorem 2.16 in [1, p. 286]. This is an open question in the source, not an affirmative conjecture that all these properties hold. We prove the full positive two-blade statement and give three-blade failures of log-concavity. We do not resolve all parts of that question.

2 Direct counting and the polynomial formula

Lemma 1. *For the graph in* (1)*,*

$$D_G(x) = \big((1+x)^r - 1\big)(1+x)^M + \prod_{i=1}^{t}\big((1+x)^{a_i} - 1\big). \tag{2}$$

Proof. Partition dominating sets by whether they meet the central clique. If they do, any nonempty subset of the central clique and any subset of the other M vertices dominate the whole graph: a selected central vertex is adjacent to every other vertex. These sets contribute $((1+x)^r - 1)(1+x)^M$.

If no central vertex is selected, every blade must contribute a nonempty subset, because vertices in a blade have no neighbors in any other blade. Conversely, nonempty selections from every blade dominate their own cliques and all central vertices. Here $t \geq 1$, so there is a selected neighbor for the central vertices. The contribution is the product in (2). The two classes are disjoint and exhaustive, proving the formula. □

Formula (2) agrees with Theorem 2.16 of [1]: its first two terms combine as

$$\big((1+x)^r - 1\big)\big((1+x)^M - 1\big) + \big((1+x)^r - 1\big) = \big((1+x)^r - 1\big)(1+x)^M.$$

The printed display in Corollary 2.17 contains an additional $((1+x)^r - 1)$ after this combined term, when its central-clique parameter is renamed r. That display is not used here. All counts follow the graph definition and Lemma 1, consistent with Theorem 2.16. This comparison does not assert an author-issued correction to the source.

With two blades of orders a, b, expanding (2) gives

$$D_G(x) = (1+x)^N - (1+x)^a - (1+x)^b + 1, \qquad N = r + a + b. \tag{3}$$

Thus

$$d_0 = 0, \qquad d_k = \binom{N}{k} - \binom{a}{k} - \binom{b}{k} \quad (1 \leq k \leq N). \tag{4}$$

We use $\binom{a}{k} = 0$ for integers $k > a$, and likewise for b. For each $1 \leq k \leq N$, fixing any one central vertex and choosing $k-1$ of the other vertices yields dominating sets, so

$$d_k \geq \binom{N-1}{k-1} > 0. \tag{5}$$

3 Ultra-log-concavity for every two-blade join

Theorem 1. *For every triple of positive integers r, a, b, the domination coefficient sequence of $K_r \vee (K_a \sqcup K_b)$ is ultra-log-concave of order $N = r + a + b$. It is therefore log-concave and unimodal.*

Proof. Because $r, a, b \geq 1$, we have $N \geq 3$ and $N - a = r + b \geq 2$, $N - b = r + a \geq 2$. Define, for $0 \leq k \leq N$,

$$A_k = \frac{\binom{a}{k}}{\binom{N}{k}}, \qquad B_k = \frac{\binom{b}{k}}{\binom{N}{k}}.$$

The denominator is strictly positive throughout this index range. By (4), the normalized coefficients satisfy

$$u_0 = 0, \qquad u_k = 1 - A_k - B_k \quad (1 \leq k \leq N). \tag{6}$$

In particular, the latter expression is not to be substituted at $k = 0$, where $1 - A_0 - B_0 = -1$ rather than u_0.

We claim that, for every $0 \le k \le N-2$,

$$A_k - 2A_{k+1} + A_{k+2} = A_k \frac{(N-a)(N-a-1)}{(N-k)(N-k-1)} \ge 0. \tag{7}$$

For $k \le a$, the two needed ratios are

$$A_{k+1} = A_k \frac{a-k}{N-k}, \qquad A_{k+2} = A_k \frac{(a-k)(a-k-1)}{(N-k)(N-k-1)}.$$

These equalities include the zero-factor cases $k = a-1$ and $k = a$; they do not divide by a vanishing numerator or by A_{k+1}. All denominators are positive because $k \le N-2$. Writing $s = N-k$ and $d = N-a$, their common numerator simplifies by

$$s(s-1) - 2(s-d)(s-1) + (s-d)(s-d-1) = d(d-1),$$

which proves (7) in this range. For $k > a$, all three terms A_k, A_{k+1}, A_{k+2} are zero by the binomial convention, and the same identity holds. This covers every boundary case. Replacing a by b proves the identical statement for B.

For $2 \le k \le N-1$, all three indices $k-1, k, k+1$ are positive, so (6) and the second-difference identities at index $k-1$ give

$$u_{k-1} - 2u_k + u_{k+1} \le 0.$$

The entries $u_1, \dots, u_N$ are positive by (5). The arithmetic-geometric mean inequality therefore gives

$$u_k \ge \frac{u_{k-1} + u_{k+1}}{2} \ge \sqrt{u_{k-1}u_{k+1}} \qquad (2 \le k \le N-1). \tag{8}$$

For the remaining index $k = 1$, the required log-concavity inequality is automatic because $u_0 = 0$: $u_1^2 \ge 0 = u_0 u_2$. Thus $(u_k)_{k=0}^N$ satisfies every required log-concavity inequality, and its positive support is $1, \dots, N$. This proves ultra-log-concavity of order N.

Finally the binomial row is positive and log-concave, since

$$\frac{\binom{N}{k}^2}{\binom{N}{k-1}\binom{N}{k+1}} = \frac{(k+1)(N-k+1)}{k(N-k)} \ge 1 \qquad (1 \le k \le N-1).$$

Multiplying its log-concavity inequality by that of u proves $d_k^2 \ge d_{k-1}d_{k+1}$ for $1 \le k \le N-1$. Since $d_1, \dots, d_N > 0$, their successive ratios are nonincreasing and can pass from at least one to at most one only once. Hence the sequence is unimodal, with any equality plateau allowed; the leading $d_0 = 0$ does not change that conclusion. □

At the smallest parameters $r = a = b = 1$, formula (3) gives $(d_0, d_1, d_2, d_3) = (0, 1, 3, 1)$ and $(u_0, u_1, u_2, u_3) = (0, 1/3, 1, 1)$. This illustrates the need to treat $k = 1$ through $u_0 = 0$: the concavity argument concerns only $u_1, \dots, u_N$, not the entire normalized sequence starting at zero.

4 Three-blade failures of ordinary log-concavity

Proposition 1. *The graph $G = K_1 \vee (K_4 \sqcup K_5 \sqcup K_7)$ has 17 vertices and a domination polynomial that is not log-concave.*

Proof. A dominating set of size one or two must contain the central vertex: otherwise each of the three blades would need a selected vertex. Thus $d_1 = 1$ and $d_2 = 16$. A dominating set of size three either contains the central vertex and two arbitrary other vertices, or contains one vertex from each blade. These disjoint cases give

$$d_3 = \binom{16}{2} + 4 \cdot 5 \cdot 7 = 120 + 140 = 260.$$

Consequently $d_2^2 - d_1 d_3 = 256 - 260 = -4 < 0$, which violates log-concavity at index two. □

This counterexample has distinct blade orders, so it does not depend on an interpretation allowing equal orders in the source's wording. It is also *unimodal*. For completeness, its full coefficient sequence, equivalently obtained by expanding

$$x(1+x)^{16} + \big((1+x)^4 - 1\big)\big((1+x)^5 - 1\big)\big((1+x)^7 - 1\big),$$

is

$$\begin{aligned}(d_0, \ldots, d_{17}) = (&0, 1, 16, 260, 1470, 4830, 10913, 18291, 23641,\\ &24034, 19371, 12363, 6187, 2380, 680, 136, 17, 1).\end{aligned}$$

It increases up to $d_9 = 24034$ and then decreases. These full coefficients also occur in the earlier exact graph enumeration; the direct three-coefficient proof above does not require that enumeration.

Proposition 2. *When equal blade orders are allowed, every graph $K_1 \vee (K_q \sqcup K_q \sqcup K_q)$ with integer $q \geq 5$ has a domination polynomial that is not log-concave.*

Proof. The same partition of sets of sizes one, two and three gives

$$d_1 = 1, \qquad d_2 = 3q, \qquad d_3 = \binom{3q}{2} + q^3.$$

Their log-concavity difference is exactly

$$d_2^2 - d_1 d_3 = -\frac{q}{2}(2q^2 - 9q - 3). \tag{9}$$

The factor $2q^2 - 9q - 3$ is 2 at $q = 5$, and its increment when q increases to $q+1$ is $4q - 7 > 0$ for $q \geq 5$. Thus (9) is strictly negative for every integer $q \geq 5$. At $q = 5$ the first three coefficients are $(1, 15, 230)$, giving difference -5. □

The equal-blade family is stated for the graph definition (1), which explicitly permits repeated orders. Its negative conclusion concerns log-concavity; no claim that this family fails unimodality is made.

5 Evidence and limits of the conclusions

All parameter-wide results above follow from the written counting and algebraic proofs. Earlier exact checks covered all 1728 ordered triples $r, a, b \in \{1, \ldots, 12\}$, including 63936 second-difference equalities and 31968 ultra-log-concavity inequalities. Separate earlier vertex-subset enumeration checked four small two-blade graphs and the full coefficient vectors of the two displayed three-blade examples. These are historical finite checks, not new executions for this note or substitutes for the general proof. Their precise execution scope is recorded in the accompanying reproducibility note; no program is needed for the mathematical argument and no larger suite is bundled.

Theorem 1 covers exactly positive r, a, b and two blades. The three-blade counterexamples rule out universal log-concavity, but do not resolve unimodality for arbitrary $t \geq 3$ or the source's general questions about zeros and their limits. No claim of smallest counterexample order, first discovery or worldwide priority is made. Prepared with AI assistance; no proof-assistant certification or independent human peer review is asserted.

References

[1] Bilal Ahmad Rather, *On domination polynomials of some graphs*, Journal of Combinatorial Mathematics and Combinatorial Computing **126** (2025), 279–289. doi:10.61091/jcmcc126-19. Theorem 2.16, its following open question, and Corollary 2.17, printed p. 286.